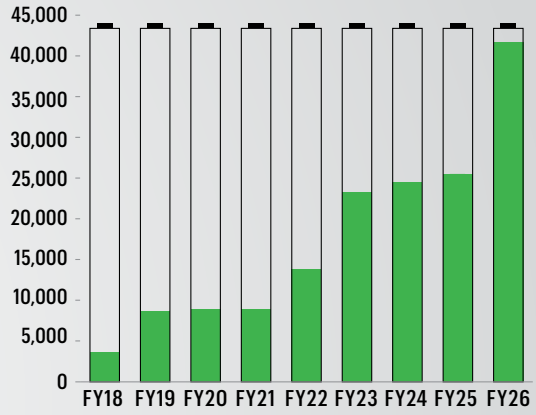
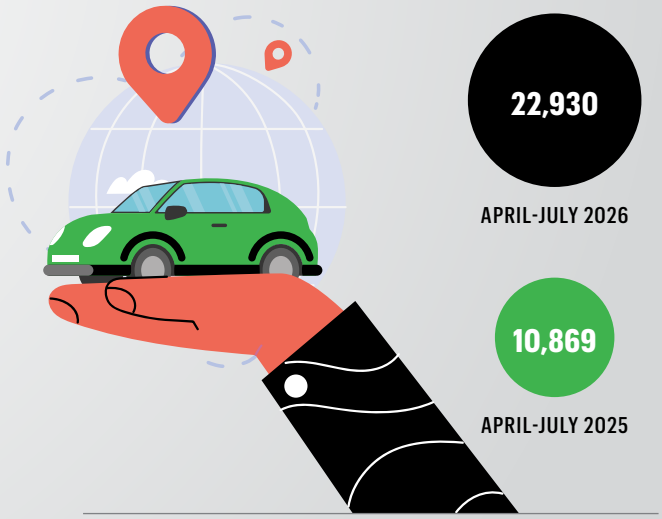
IMPORTS SURGE

• Rising demand for EVs has led to a sharp rise in lithium-ion cell imports



Cell imports in FY27 versus FY26



₹ CR; **SOURCE:** COMMERCE MINISTRY

anode account for 65% of a cell's cost," says Choudhary. Himadri plans to invest ₹4,200 crore for building cathode active materials; lithium carbonate comprises 25% of such materials. Since lithium is not mined in India, Himadri plans to buy from Argentina, Bolivia and Chile.

Electrolyte is the only component that cell producers buy locally, while everything else comes from China, as no Indian company produces raw materials at scale, says Vikram Handa, founder and managing director of Epsilon Advanced Materials. Over 90% of cathode and anode active material is processed in China, says Handa.

The government has been trying to fix things, but the industry is struggling, says Handa. "I think the intention of ACC PLI was that demand is coming and so let's promote this segment. In the last three-four years, the challenge has been that it is a high technology space that is moving extremely fast," he says.

Lack of investment in technology is one reason for the situation, says Handa. "The biggest challenge is risk capital. Today, someone putting in ₹2,000 crore wants to earn in the next quarter. That's difficult in this industry. Are companies ready to invest 8-10% on R&D? Not many are. The government can only nudge and bring in policy enablers," he says.

The key is to look at the entire battery value chain rather than each part in isolation, says Handa. "If we want to localise cell manufacturing, we also need to localise the materials and components that go into the cell. Otherwise, we will continue to depend on imports for a large part of the value chain, which can also become a national security issue," says Handa.

Epsilon makes graphite anodes at its 2,000-tonne plant, which can cater to 2 GWh of cell capacity. It is looking to scale it up to 30,000 tonnes by 2028. Epsilon's anode material project has secured approval under MeitY's Electronics Component Manufacturing Scheme with ₹145 crore in capex support, covering 25% of the capital investment.

Battery cells produced in India are expected to cost roughly 20% more than Chinese imports, says Himadri's Choudhary. "China sells below cost so that no one finds it commercially viable," he says. To make India self-reliant in battery cells, both operational and capital incentives are needed, says Choudhary. "We need support for the initial years," he says.

"There is a need for government support for cell components. China sells at a very competitive price. So, there are hardly any margins. The Indian government has to come up with a component scheme to make the business viable," says Choudhary.

An executive of a cell company in India told Epsilon's Handa that if they buy everything from China at the cheapest price and not buy anything locally, their cell is still 25% more expensive than a Chinese cell.

MAKING A START

Lead acid battery maker Amara Raja Energy & Mobility, which is building the technical know-how in-house after its proposed partnership with China's Gotion hit a wall, is slowly warming up to locally made cell components for the recently commissioned Customer Qualification Plant (CQP) at its Giga Corridor in Telangana. With an initial capacity of 60 MWh, it will make lithium-ion cells that customers can test and validate ahead of commercial production.

"We use mainly imported materials today. But some Indian materials are being made available. We have started making batches of cells with select